



Digital Signal Processing

(EE3031)

Date: May 27th 2025

Course Instructor(s)

Dr. Muhammad Akmal

Final Exam

Total Time (Hrs): 3

Total Marks: 50

Total Questions: 5

Roll No

Section

Student Signature

Attach the question paper with answer sheet.
Do not write below this line.

Attempt all the questions.

CLO # 5: Design filters using various techniques

Q1:

[10]

A digital low-pass filter is required to meet the following specifications:

Passband ripple: ≤ 2 dB

Passband edge: 6 kHz

Stopband attenuation: ≥ 50 dB

Stopband edge: 9 kHz

Sample rate: 34 kHz

The filter is to be designed by performing a bilinear transformation on an analog system function. Determine what order i) Butterworth, and ii) Chebyshev I analog designs must be used to meet the specifications in the digital implementation.

CLO # 3: Apply z-transform to analyze discrete time systems

Q2: Determine the z-transform of the following signals:

[10]

a) $x(n) = a^n (\cos w_0 n) u(n)$

b) $x(n) = a^n (\sin w_0 n) u(n)$

CLO # 5: Design filters using various techniques

Q3:

[10]

Design an FIR Bandstop digital filter approximating the following frequency response:

$$H_d(\omega) = \begin{cases} 1 & \text{for } |\omega| \leq \frac{\pi}{6} \\ 0 & \text{for } \frac{\pi}{6} < |\omega| \leq \frac{\pi}{3} \\ 1 & \text{for } \frac{\pi}{3} \leq |\omega| \leq \pi \end{cases}$$



- a) Determine the coefficients of a 5-tap filter based on the window method with a i) Hanning window, and ii) Blackman window.
(b) Determine and plot the magnitude, and phase response of the filter.

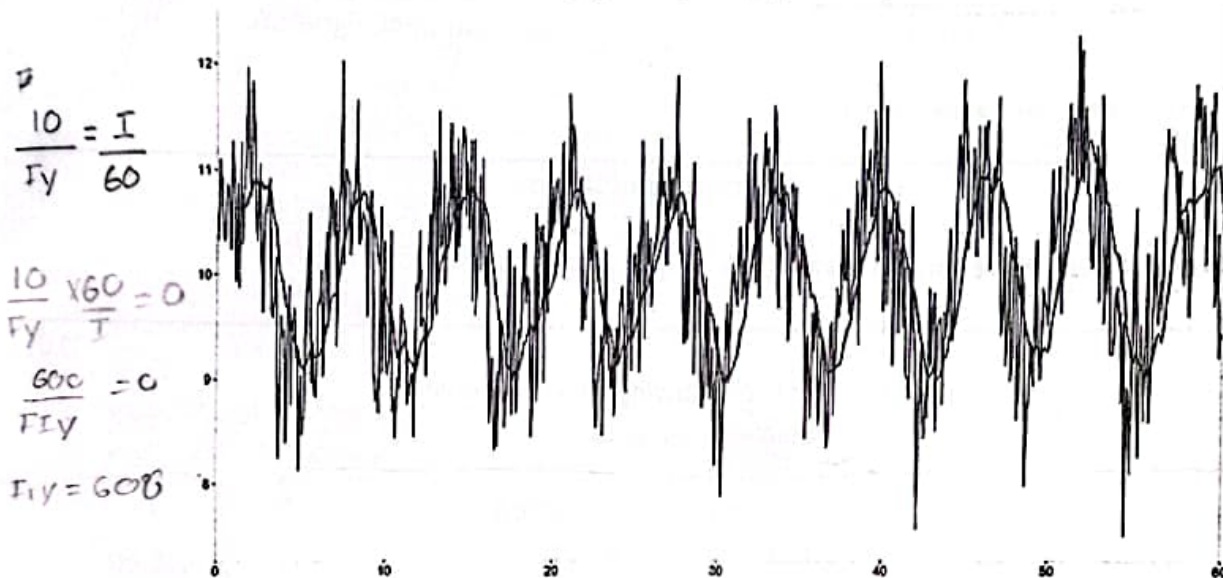
CLO # 3: Apply z-transform to analyze discrete time systems

Q4:

[10]

An audio processing system samples an analog signal at 10 kHz, and uses a digital filter to remove 60 Hz power-line noise. The input-output relation of the system is as follows:

$$y[n] = 0.8y[n-1] + x[n] - 0.9x[n-1]$$



where:

$x[n]$ = input signal

$y[n]$ = filtered output

Determine the transfer function $H(z)$ of the filter using the Z-transform. Also show magnitude, and Pole-Zero locations.

CLO # 5: Design filters using various techniques

Q5:

[10]

Design a type I Low Pass Chebyshev filter that has a 1-dB ripple in the passband, a cutoff frequency $\Omega_p = 1000\pi$, a stopband frequency of 2000π , and an attenuation of 40dB or more for $\Omega \geq \Omega_s$.